

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Information Security	Course Code:	CS3002
	Program:	BCS	Semester:	Fall 2026
	Sections:	7H,7K	Total Marks:	10
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	Exam Type:	Assignment 2	Page(s):	2

Block Cipher Modes Decryption

AES is a block cipher that operates on fixed 128-bit (16-byte) blocks. To encrypt messages longer than one block, AES must be combined with a mode of operation, which defines how successive blocks are processed and/or chained together.

You studied two common modes: Cipher Block Chaining (CBC) and Counter (CTR).

For this assignment, you are provided with two ciphertexts, each encrypted with AES-128 using a different mode of operation. For each, you are also given the corresponding key and IV/counter. Your task is to decrypt both ciphertexts block by block, recovering the original plaintext, and to document the process using diagrams that trace every intermediate value through the chain.

Provided Data

All values below are given in hexadecimal.

Message 1: AES-128-CBC

Ciphertext (3 blocks)	9a8460ce302b68a3feb172a2fd89a5b2 8f203e5e67ad1bc5022fa2837a2b8222 77cc0976ac42593dc8b5b4e70816ec87
IV	ad8331c53199ac271845afaf43066668
Key	2667172251915cc44cfc2ec3a88b4d89

Message 2: AES-128-CTR

Ciphertext (3 blocks)	7285090a261f8fb3a3b69b2cbb9cf940 59d2fb7cfe3fb40e264bc0dec8c90d1a 0f3788c2e2a2
Counter	d441e7c1fa329130bbaea83918ea5cc9
Key	53153913a1623f8a089ee17360fb9ac9

In addition, a web-based tool <https://pkitools.net/> is provided to you. It contains a calculator for one AES block operation (encryption or decryption). It also includes a XOR calculator. Use this tool repeatedly, using its outputs by hand according to the rules of each mode, to decrypt the full cipher texts. (You can use any other online available calculator)

Tasks

Part A. CBC Mode

1. Draw a diagram of the CBC decryption process for the given three-block ciphertext.
2. On the diagram itself, write in the actual hexadecimal values at every point. This includes each ciphertext block, the IV, the key, the output of each AES decryption operation, each XOR result, and each recovered plaintext block.
3. Using the AES Lab tool, carry out the decryption block by block and fill in these values in the above diagram as you go.
4. You learnt in lectures that CBC encryption is inherently sequential. Each block depends on the ciphertext of the previous one, so it cannot be parallelized. Does the same limitation apply to CBC decryption? Justify your answer using your diagram.

Part B. CTR Mode

5. Repeat Tasks 1–3 for the CTR-encrypted message: draw a diagram with all actual values filled in (counter values for each block, AES output, ciphertext, XOR result, plaintext), and use the AES Lab tool to recover the plaintext block by block.

Part C

6. After decrypting both messages, you may notice that the CBC-recovered plaintext includes trailing padding (zero bytes), while the CTR-recovered plaintext does not. Explain why padding is not required in CTR mode, in terms of how the mode processes data.

Deliverables

A single document containing

- Two fully filled diagrams (CBC and CTR decryption)
- The recovered plaintext for both messages
- Written answers to the discussion questions 4 and 6